12TH CHEMISTRY · CHAPTER 19 · FULL CHAPTER NOTES

Basics of Organic Chemistry

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

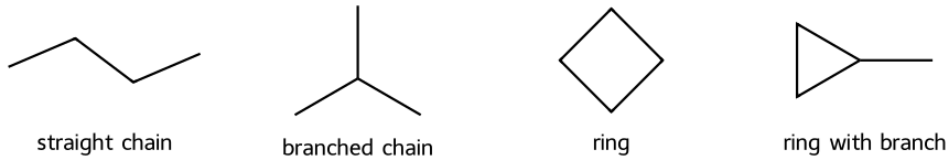
Q.1 [Short] Define organic chemistry. Which carbon compounds are excluded from it? How do organic compounds differ from inorganic compounds?

IUPAC defines organic chemistry as “the study of compounds containing carbon, where carbon is bonded to itself, hydrogen, or other elements excluding some inorganic compounds.” The traditional inorganic compounds excluded are carbonates, cyanides (e.g., KCN), CO_2, CO, etc. Organic compounds are easily distinguished from inorganic compounds by their low melting points, low boiling points, low densities and low solubility in water. More than 20 million carbon-containing compounds are known today.

19.1 CATENATION

Q.2 [Short] Define catenation and explain its importance in organic chemistry.

The ability of self-linkage between the atoms of the same element is known as **catenation** (from the Latin *catena*, chain). Carbon forms long chains and rings through stable C–C covalent bonds. The tetravalent nature of carbon gives it many different ways of arrangement; for example, four carbon atoms can be arranged in four different ways. **Importance:** catenation explains why organic compounds are complex, have large structures and show vast diversity.

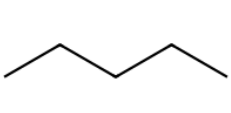


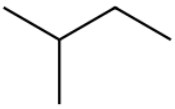
Four different arrangements of four carbon atoms

Quick Check 19.1

Q.3 [Short] a) Draw the different possible structures containing five carbon atoms.

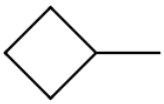
Five carbon atoms can be joined as three open chains and five rings (carbon skeletons shown as bond-line structures):

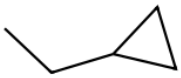

pentane


2-methylbutane


2,2-dimethylpropane


cyclopentane


methylcyclobutane


ethylcyclopropane

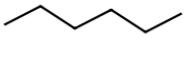

1,1-dimethylcyclopropane

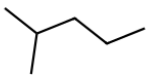

1,2-dimethylcyclopropane

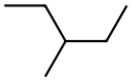
Open chains: pentane, 2-methylbutane, 2,2-dimethylpropane. Rings: cyclopentane, methylcyclobutane, ethylcyclopropane, 1,1- and 1,2-dimethylcyclopropane.

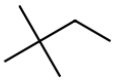
Q.4 [Short] b) Give different open chain structures containing six carbon atoms.

There are five open chain arrangements of six carbon atoms:


hexane


2-methylpentane


3-methylpentane


2,2-dimethylbutane


2,3-dimethylbutane

They are hexane, 2-methylpentane, 3-methylpentane, 2,2-dimethylbutane and 2,3-dimethylbutane.

19.2 HYDROCARBONS (COMPOUNDS OF CARBON AND HYDROGEN ONLY)

Q.5 [Long] What are hydrocarbons? Classify them with examples.

Compounds containing carbon and hydrogen only in their molecules are called **hydrocarbons**. On the basis of the carbon skeleton they are of two types:

1. **Open chain (acyclic) or aliphatic hydrocarbons:** have open chains of carbon atoms (*aleiphatos* is a Greek word meaning fat; fats have open chain structures). They are subdivided into saturated hydrocarbons (alkanes) and unsaturated hydrocarbons (alkenes and alkynes). Examples: ethane CH_3–CH_3, ethene $H_2C{=}CH_2$, ethyne HC≡CH.
2. **Closed chain or cyclic hydrocarbons:** have rings of carbon atoms. (i) **Alicyclic** hydrocarbons are cyclic hydrocarbons, saturated or unsaturated, that behave like aliphatic hydrocarbons, e.g., cycloalkanes, cycloalkenes (cyclohexane). (ii) **Aromatic** hydrocarbons have a benzene ring or resembling structures, e.g., benzene, methylbenzene, naphthalene, anthracene, diphenylmethane.

Q.6 [Short] Differentiate between alicyclic and aromatic hydrocarbons.

Alicyclic hydrocarbons	Aromatic hydrocarbons
Cyclic hydrocarbons (saturated or unsaturated) that behave like aliphatic hydrocarbons.	Cyclic hydrocarbons having a benzene ring or resembling structures.
Examples: cycloalkanes, cycloalkenes, e.g., cyclohexane.	Examples: benzene, methylbenzene, naphthalene, anthracene.

Q.7 [Short] Differentiate between saturated and unsaturated hydrocarbons.

Saturated hydrocarbons (alkanes) contain only carbon–carbon single bonds, e.g., ethane CH_3–CH_3. Unsaturated hydrocarbons contain a carbon–carbon double bond (alkenes, e.g., ethene $H_2C{=}CH_2$) or triple bond (alkynes, e.g., ethyne HC≡CH).

Quick Check 19.2

Q.8 [Short] a) Classify the following compounds as an aliphatic, alicyclic or aromatic.

No.	Structure given	Class
i)	Cyclohexane ring carrying a CH_3 group (methylcyclohexane)	Alicyclic
ii)	Six-membered ring with two double bonds carrying a CH_3 group	Alicyclic
iii)	Benzene ring carrying a CH_3 group (methylbenzene)	Aromatic
iv)	Two fused benzene rings (naphthalene)	Aromatic
vii)	Cyclohexane	Alicyclic

No structures are printed for parts v) and vi) in the textbook, so they cannot be classified. None of the printed structures is aliphatic (open chain).

19.3 FUNCTIONAL GROUPS

Q.9 [Short] Define functional group.

An atom or group of atoms, or the presence of a double or triple bond, which determines the characteristic properties of an organic compound is called a **functional group**. Both the physical and the chemical properties of organic compounds are highly affected by the nature of the functional group they possess.

Q.10 [Short] How do functional groups affect the physical properties of organic compounds? Give examples.

- Methanol and ethanol are liquids, while methane and ethane are gases.
- The melting and boiling points of alcohols are higher than those of the corresponding alkenes. This is due to the hydroxyl group (–OH), which can form hydrogen bonds.

Alkanes lack this ability as they have no such functional group.

- Lower alcohols and carboxylic acids, glucose and sucrose are soluble in polar solvents like water due to their polar functional groups.

Q.11 [Short] Why do alkanes and cycloalkanes have no functional group?

The type of an organic compound is dictated by two factors in its functional group: (i) the polarity of the bond and (ii) the bond order, i.e., the number of bonds. Alkanes (and cycloalkanes) contain only non-polar C–C and C–H single bonds. They possess no charges or polarity and no double or triple bonds in their structures, so they have no functional group.

19.4 NOMENCLATURE OF ALIPHATIC ORGANIC COMPOUNDS

Q.12 [Short] Why was the IUPAC system of nomenclature needed?

Millions of organic compounds are found in nature, and hundreds and thousands of new compounds are synthesised every year. A need was felt to name these compounds according to their structures. So IUPAC (International Union of Pure and Applied Chemistry) formulated a set of rules for the systematic nomenclature of organic compounds.

Q.13 [Long] Explain the components of the IUPAC name of an organic compound with examples.

The systematic (IUPAC) name of an organic compound is based on the parent hydrocarbon. Each name consists of the following components:

1. **Stem or root:** gives the longest continuous carbon chain including the carbon bearing the functional group. The lowest possible number is given to the carbon bearing the functional group. If the carbon atoms are joined in a ring, the name starts with “cyclo”.
2. **Suffix:** (a) the **primary suffix** (suffix of the stem) shows whether the parent chain is saturated or unsaturated (an, en, yn); (b) the **secondary suffix** (suffix of the key functional group) gives the family name of the compound.
3. **Prefix:** gives the name of the side chain (alkyl group) or any substituent (e.g., halo, nitro) along with the number of the carbon atom to which it is attached. If the prefix is absent, the name consists of root and suffix only.

Compound	Prefix	Stem	Suffix of stem	Suffix of F.G.	Name
$Cl–CH_2–CH_2–CH_2–CHO$	4-chloro	but	an	al	4-chlorobutanal
$CH_3–C(CH_3)=CH–CH_2–OH$	3-methyl	but	2-en	1-ol	3-methylbut-2-en-1-ol

In the textbook’s first example the name is printed as “4-chloro-prop-an-al”; the chain drawn has four carbon atoms, so the stem is “but” (4-chlorobutanal).

Q.14 [Long] **Give the general formula, functional group and IUPAC naming of the different classes of organic compounds with examples.**

#	Class	General formula	Functional group	IUPAC name	Examples
1	Alkyl halides	$C_nH_{2n+1}X$	–C–X	Halogenoalkane	CH_3Cl chloromethane; $CH_3CH(Br)CH_3$ 2-bromopropane
2	Alcohols	$C_nH_{2n+2}O$	–C–OH	Alkanol	CH_3OH methanol; $CH_3CH_2CH_2OH$ propan-1-ol; $(CH_3)_3COH$ 2-methylpropan-2-ol
3	Aldehydes	$C_nH_{2n}O$	–CHO	Alkanal	HCHO methanal; CH_3CHO ethanal; $(CH_3)_2CHCH_2CHO$ 3-methylbutanal
4	Ketones	$C_nH_{2n}O$	C–CO–C	Alkanone	CH_3COCH_3 propanone; $CH_3CH_2CH_2COCH_3$ pentan-2-one
5	Carboxylic acids	$C_nH_{2n}O_2$	–COOH	Alkanoic acid	HCOOH methanoic acid; CH_3COOH ethanoic acid; $CH_3CH_2CH(CH_3)COOH$ 2-methylbutanoic acid
6	Acid amides	$C_nH_{2n-1}ONH_2$	$–CONH_2$	Alkanamide	CH_3CONH_2 ethanamide; $(CH_3)_2CHCONH_2$ 2-methylpropanamide; $CH_3CH_2CH_2CONHCH_3$ N-methylbutanamide
7	Acid halides	$C_nH_{2n-1}OX$	–COX	Alkanoyl halide	CH_3COCl ethanoyl chloride; CH_3CH_2COCl propanoyl chloride
8	Esters	$C_nH_{2n}O_2$	–COO–C	Alkyl alkanoate	CH_3COOCH_3 methyl ethanoate; $CH_3COOCH_2CH_3$ ethyl ethanoate; $CH_3CH_2COOCH_2CH_2CH_3$ propyl propanoate
9	Acid anhydrides	—	–CO–O–CO–	Alkanoic anhydride	$(CH_3CO)_2O$ ethanoic anhydride; $CH_3CO–O–COCH_2CH_3$ ethanoic propanoic anhydride

#	Class	General formula	Functional group	IUPAC name	Examples
10	Ethers	$C_nH_{2n+2}O$	C–O–C	Alkoxyalkane	CH_3OCH_3 methoxymethane; $CH_3CH_2OCH_3$ methoxyethane; $CH_3CH_2OCH_2CH_3$ ethoxyethane; $(CH_3)_2CHCH_2OCH_3$ 1-methoxy-2-methylpropane
11	Amines	$C_nH_{2n+1}NH_2$	$–NH_2$, –NH–R, $–NR_2$	Alkanamine	see the next table
12	Alkyl cyanides	$C_nH_{2n–1}N$	–C≡N	Alkanenitrile	CH_3CN ethanenitrile; $(CH_3)_2CHCN$ 2-methylpropanenitrile
13	Nitroalkanes	$C_nH_{2n+1}NO_2$	$–NO_2$	Nitroalkane	CH_3NO_2 nitromethane; $CH_3CH_2CH_2NO_2$ nitropropane; $CH_3CH(NO_2)CH_3$ 2-nitropropane

Amines are of three types:

Type	Group	IUPAC name	Examples
Primary	$–C–NH_2$	Alkanamine	$CH_3CH_2NH_2$ ethanamine; $CH_3CH_2CH_2NH_2$ propan-1-amine; $CH_3CH(NH_2)CH_3$ propan-2-amine
Secondary	–C–NH–R	N-Alkylalkanamine	$CH_3CH_2CH_2NHCH_3$ N-methylpropan-1-amine; $(CH_3)_2CHNHCH_3$ N-methylpropan-2-amine
Tertiary	$–C–NR_2$	N,N-Dialkylalkanamine (R = R); N-alkyl-N-alkyl alkanamine (R ≠ R)	$CH_3CH_2CH_2N(CH_3)_2$ N,N-dimethylpropan-1-amine; $CH_3CH_2CH_2N(CH_3)CH_2CH_3$ N-ethyl-N-methylpropan-1-amine

Q.15 [Short] Differentiate between primary, secondary and tertiary amines.

In a primary amine the nitrogen is bonded to one alkyl group ($–NH_2$), e.g., ethanamine. In a secondary amine it is bonded to two alkyl groups (–NH–R), e.g., N-methylpropan-1-amine. In a tertiary amine it is bonded to three alkyl groups ($–NR_2$), e.g., N,N-dimethylpropan-1-amine.

Q.16 [Short] Aldehydes and ketones have the same general formula. How do they differ?

Both have the general formula $C_nH_{2n}O$ and contain the carbonyl group. In aldehydes (alkanals) the carbonyl carbon is bonded to at least one hydrogen atom (–CHO), e.g., ethanal CH_3CHO. In ketones (alkanones) it is bonded to two carbon atoms, e.g., propanone CH_3COCH_3.

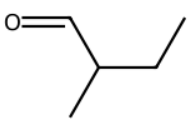
Quick Check 19.3

Q.17 [Short] a) What are the systematic names of the following?

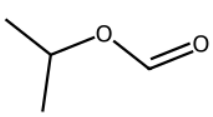
No.	Formula	Systematic name
i)	$(CH_3)_3COH$	2-methylpropan-2-ol
ii)	CH_3CCl_3	1,1,1-trichloroethane
iii)	$CH_2=CH–CH_2–OH$	prop-2-en-1-ol
iv)	$CH_3COC_2H_5$	butan-2-one (butanone)
v)	$CH_2=C(CH_3)COOCH_3$	methyl 2-methylprop-2-enoate

Q.18 [Short] b) Draw the structures of the following.

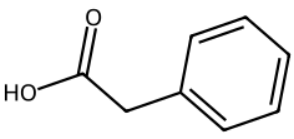
No.	Name	Condensed structure
i)	2-methylbutanal	$CH_3–CH_2–CH(CH_3)–CHO$
ii)	prop-2-ylmethanoate	$HCOO–CH(CH_3)_2$
iii)	phenylethanoic acid	$C_6H_5–CH_2–COOH$
iv)	cyclohexanone	six-membered carbon ring with one C=O; $C_6H_{10}O$
v)	3-methylpentanal	$CH_3–CH_2–CH(CH_3)–CH_2–CHO$



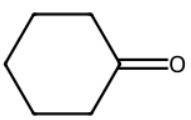
i) 2-methylbutanal



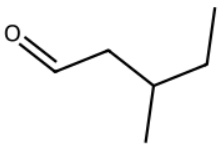
ii) prop-2-yl methanoate



iii) phenylethanoic acid



iv) cyclohexanone



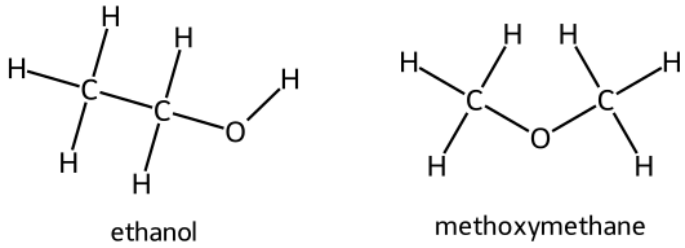
v) 3-methylpentanal

The skeletal structures above show the same five compounds.

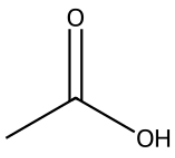
19.5 REPRESENTING ORGANIC MOLECULES

Q.19 [Long] **Describe the different types of formulae used to represent organic compounds.**

Formula	What it shows	Examples
Molecular formula	The actual number of atoms of each element present in a molecule of a compound. It gives no information about the sequence of atoms or the functional group.	butane C_4H_{10}; acetic acid $C_2H_4O_2$; ethanol C_2H_6O
Empirical formula	The smallest whole number ratio of atoms of each element present in a molecule.	butane C_2H_5; ethanoic acid CH_2O
Structural formula	The number of each kind of atom and also the sequence in which the atoms are joined together.	displayed, condensed and skeletal formulae
Displayed (graphical) formula	All the bonds between the different atoms, including the type of bonds and the positions of functional groups.	every C–H, C–C, C–O bond drawn
Condensed (linear) formula	A one-line structure: single covalent bonds are omitted; branches, functional groups, double or triple bonds and repeated methylene groups are shown (in brackets).	butane $CH_3(CH_2)_2CH_3$; ethanol CH_3CH_2OH; ethanoic acid CH_3COOH
Skeletal (bond-line) formula	Symbols of C and H atoms and the C–H bonds are removed. C–C bonds are single lines meeting at an angle; each carbon is assumed to form enough C–H bonds to make four bonds. Multiple bonds and functional groups are shown.	zig-zag lines; used for more complex molecules



Displayed formulae



Skeletal formula of ethanoic acid

Q.20 [Short] Why can two different compounds have the same molecular formula? Give an example.

A molecular formula gives no information about the sequence in which the atoms are linked or about the nature of the functional group. So ethanol (CH_3CH_2OH) and methoxymethane (CH_3OCH_3) are both represented by C_2H_6O.

Q.21 [Short] Differentiate between molecular formula and empirical formula.

Molecular formula	Empirical formula
Shows the actual number of atoms of each element in a molecule.	Shows the smallest whole number ratio of atoms of each element.
Butane C_4H_{10}; ethanoic acid $C_2H_4O_2$.	Butane C_2H_5; ethanoic acid CH_2O.

Q.22 [Short] What does the term “planar” mean for organic molecules? Give examples.

A molecule or part of a molecule is planar when all its atoms lie in the same plane. The two carbon atoms of a C=C double bond and the four atoms attached to them lie in one plane (e.g., ethene, but-2-ene), because these carbon atoms are sp^2 hybridized. Benzene is also a planar molecule.

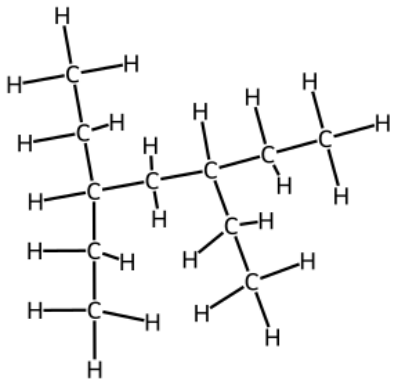
The textbook uses this idea in section 19.9 (but-2-ene); benzene is treated in Chapter 20.

Quick Check 19.4

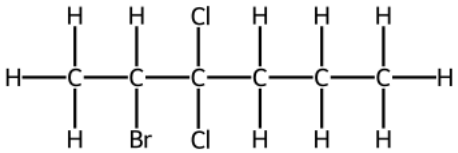
Q.23 [Short] a) Draw the displayed and skeletal formulae of the following: i. 3,5-diethylheptane ii. 2-bromo-3,3-dichlorohexane iii. 2-methylbutanal

Condensed formulae: i. $CH_3CH_2CH(C_2H_5)CH_2CH(C_2H_5)CH_2CH_3$; ii. $CH_3CH(Br)CCl_2CH_2CH_2CH_3$; iii. $CH_3CH_2CH(CH_3)CHO$.

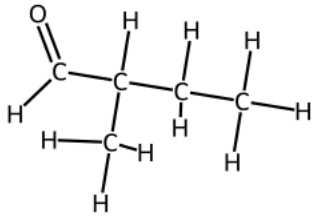
Displayed formulae (all atoms and bonds shown):



i. 3,5-diethylheptane

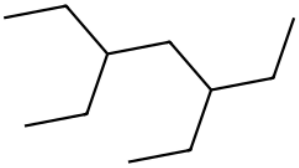


ii. 2-bromo-3,3-dichlorohexane

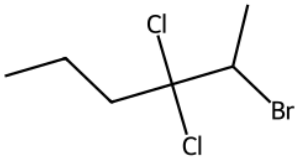


iii. 2-methylbutanal

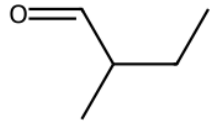
Skeletal formulae:



i. 3,5-diethylheptane



ii. 2-bromo-3,3-dichlorohexane

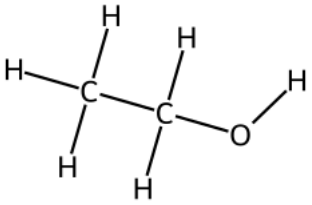


iii. 2-methylbutanal

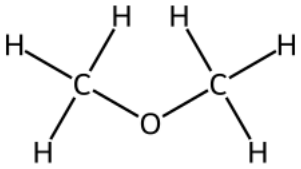
In the skeletal formulae each corner or line end is a carbon atom; hydrogen atoms on carbon are not drawn.

Q.24 [Short] **b) Draw the displayed and condensed structures of ethanol and methoxy methane, and explain whether they are isomers of the same molecule or not.**

Displayed formulae:



ethanol

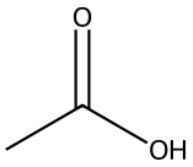


methoxymethane

Condensed formulae: ethanol CH_3CH_2OH; methoxymethane CH_3OCH_3. Both have the same molecular formula, C_2H_6O, but different structures and different functional groups (alcohol –OH and ether C–O–C). So they are isomers of each other (functional group isomers), not the same compound.

Q.25 [Short] **c) Draw the skeletal formula of ethanoic acid (CH_3COOH).**

The line end is the CH_3 carbon and the corner is the carboxyl carbon; the C=O and O–H groups are shown.



19.6 DETERMINATION OF THE MOLECULAR FORMULA

Q.26 [Short] **Write the steps to find the molecular formula from a skeletal formula.**

1. Count the C atoms represented and the heteroatoms shown in the skeleton.
2. Each line represents a bond and each joint shows a carbon atom (double lines represent double bonds). Count one C atom at the end of each line unless another atom is shown there.
3. Count the number of H atoms necessary to complete the octet (four bonds) of each C atom.

Q.27 [Short] **Find the molecular and empirical formulae of (a) the skeletal structure of but-2-ene and (b) the skeletal structure of ethanamide.**

(a) Four carbon atoms: two end carbons with 3 H each and two doubly bonded carbons with 1 H each. Molecular formula C_4H_8; empirical formula CH_2.

(b) Two carbon atoms: the end carbon has 3 H, the C=O carbon has no H; the NH_2 group is shown. Molecular formula C_2H_5NO; this is also its empirical formula (CH_3CONH_2).

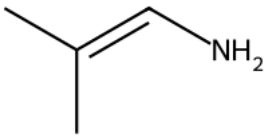
Quick Check 19.5

Q.28 [Short] **a) Write molecular formula for the compound represented by the following bond-line structure:**

The structure is Br–CH_2–CH_2–COOH (3-bromopropanoic acid): 3 C atoms, 5 H atoms, 1 Br atom and 2 O atoms. Molecular formula: **$C_3H_5BrO_2$**.

Q.29 [Short] **b) Draw the skeletal formula for $(CH_3)_2C=CHNH_2$**

The two line ends are the CH_3 groups; the C=C double bond and the NH_2 group are shown.



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Q.30 [Short] **c) Write molecular formula for the compound represented by the following bond-line structure:**

The structure is a cyclohexane ring in which one ring carbon carries a methyl group and a –CH=CH–CH_3 group. Ring: 6 C and 10 H; methyl: CH_3; side chain: C_3H_5. Molecular formula: **$C_{10}H_{18}$**.

Q.31 [Short] **d) Give the molecular formulae of the following.**

i) 1,2-dichlorocyclohexane: **$C_6H_{10}Cl_2$**. ii) Benzene ring carrying –CH_2Cl and –CO_2H groups on adjacent carbons: **$C_8H_7ClO_2$**.

In part ii) the side group is printed as "CH_3Cl"; a group attached to the ring must be –CH_2Cl.

19.7 UNDERSTANDING ORGANIC REACTIONS

A. Breaking of Covalent Bonds

Q.32 [Long] **Explain homolytic and heterolytic fission of a covalent bond.**

Homolytic fission (homolysis) occurs when the breaking bond is non-polar. The two shared electrons of the bond are distributed **equally** between the two atoms; each atom gets one electron. The atoms or groups of atoms with unpaired electrons that are formed are called **free radicals**. Free radicals are reaction intermediates and are highly reactive because of the tendency of their unpaired electrons to be paired.

$$Cl–Cl \rightarrow Cl\bullet + Cl\bullet$$

$$H_3C–H \rightarrow \bullet CH_3 + H\bullet$$

Heterolytic fission (heterolysis) occurs when the breaking bond is polar. The two shared electrons are distributed **unequally**: one atom keeps both electrons and acquires a negative charge; the other atom is deficient in one electron and acquires a positive charge. Ions are formed. A species containing a carbon atom with a positive charge is a **carbocation**; it forms when carbon is bonded to a heteroatom more electronegative than carbon (O, N, S, P, halogens). A **carbanion** forms when carbon is bonded to a heteroatom (metal) less electronegative than carbon (Li, Mg, Cd).

$$(CH_3)_3C–Cl \rightarrow (CH_3)_3C^+ + Cl^-$$

Homolytic fission	Heterolytic fission
The breaking bond is non-polar.	The breaking bond is polar.
The shared electrons are divided equally: one to each atom.	The shared electrons are divided unequally: both go to one atom.
Free radicals (neutral, with an unpaired electron) are formed.	Ions are formed: a cation (e.g., a carbocation) and an anion.
$Cl–Cl \rightarrow Cl\bullet + Cl\bullet$	$(CH_3)_3C–Cl \rightarrow (CH_3)_3C^+ + Cl^-$
Leads to free radical mechanisms.	Leads to ionic or polar mechanisms.

Q.33 [Short] What are free radicals? Why are they highly reactive?

Atoms or groups of atoms with unpaired electrons are called free radicals. They are formed when a non-polar bond is broken. They are highly reactive because of the tendency of their unpaired electrons to be paired.

Q.34 [Short] Differentiate between a carbocation and a carbanion.

Carbocation	Carbanion
A species containing a carbon atom with a positive charge.	A species containing a carbon atom with a negative charge.
Formed when carbon is bonded to a heteroatom more electronegative than carbon (O, N, S, P, Cl).	Formed when carbon is bonded to a heteroatom (metal) less electronegative than carbon (Li, Mg, Cd).
It is an electrophile.	It is a nucleophile.

B. Types of Reagents

Q.35 [Long] What are reagent and substrate? Describe the three types of reagents.

A reaction between two reactants may be regarded as the attack of one species on the other. The attacking species is called the **reagent**; the species undergoing attack is the **substrate**.

Reagent	Definition	Examples
Free radical	An atom or group of atoms having an unpaired electron.	Cl•, H•, •CH_3, CH_3COO•
Electrophile (electron lover)	An atom or group of atoms able to accept a pair of electrons to form a new bond. Positive ions (including carbocations) or neutral molecules.	CH_3^+, H^+, Br^+, NO_2^+, SO_3, BF_3
Nucleophile (nucleus lover)	An atom or group of atoms able to donate a pair of electrons to form a new bond. Negative ions (including carbanions) or neutral molecules.	CH_3^-, Br^-, OH^-, CN^-, NH_3, H_2O, ROH

The terms electrophilic and nucleophilic do not apply only to ions. Many organic compounds are polar and carry partial charges. In CH_3–OH the carbon (δ+) is electrophilic and the oxygen (δ–) is nucleophilic. In CH_3–Li the carbon (δ–) is nucleophilic and the lithium (δ+) is electrophilic.

C. Types of Organic Reactions

Q.36 [Long] Describe the main types of organic reactions with examples.

Type	Description	Example
Substitution	An atom or group of atoms in one molecule is replaced by another atom or group. Two products may be formed.	$CH_3–CH_2–Br + KOH_{(aq)} \rightarrow CH_3–CH_2–OH + KBr$
Addition	Two reactants react to produce a single product.	$CH_2=CH_2 + H_2 \rightarrow CH_3–CH_3$ (Ni, 200–300 °C); $CH_3–CH=CH_2$ + H–Br → bromopropane
Elimination	Atoms or groups are removed from two adjacent atoms of the substrate to form a multiple bond. The reverse of addition.	$CH_3–CH_2–Br$ + KOH (alcohol) → $CH_2=CH_2 + KBr + H_2O$
Hydrolysis	A compound breaks into two on reaction with water, often helped by H^+ or OH^- as catalyst.	$CH_3COOCH_2CH_3 + H_2O \rightarrow CH_3COOH + CH_3CH_2OH$ (H^+)
Condensation	Two molecules combine to form a bigger molecule, with or without loss of a smaller molecule like H_2O, NH_3, HCl.	$(CH_3)_2C=O + H_2N–OH \rightarrow (CH_3)_2C=N–OH + H_2O$ (propanone oxime)
Oxidation	Loss of electrons; gain of oxygen or loss of hydrogen; increase in the number of C–O bonds or in the oxidation state of carbon.	$CH_3–CH=CH_2 \rightarrow CH_3CHO + HCHO$ (O_3, then Zn/H_2O)
Reduction	Gain of electrons; loss of oxygen or gain of hydrogen; decrease in the number of C–O bonds or in the oxidation state of carbon.	$CH_2=CH_2 + H_2 \rightarrow CH_3–CH_3$ (Ni)

Q.37 [Short] Why are elimination reactions regarded as the reverse of addition reactions?

In addition, two reactants combine at a multiple bond to give a single saturated product. In elimination, atoms or groups are removed from two adjacent atoms to form a multiple bond. So one is the reverse of the other.

Q.38 [Short] Differentiate between oxidation and reduction in organic chemistry.

Oxidation	Reduction
Loss of electrons.	Gain of electrons.
Gain of oxygen or loss of hydrogen.	Loss of oxygen or gain of hydrogen.
Number of C–O bonds or the oxidation state of carbon increases.	Number of C–O bonds or the oxidation state of carbon decreases.

19.8 MECHANISMS OF CHEMICAL REACTIONS

Q.39 [Short] What is a reaction mechanism? Name its two main types.

A **mechanism** is the pathway adopted by a reaction, showing the breaking and making of bonds in the substrate leading to the formation of the product. Depending on the nature of the reagents, mechanisms are of two main types: (1) free radical mechanism and (2) ionic or polar mechanism.

Q.40 [Short] Describe the free radical mechanism.

The **free radical mechanism** involves homolysis, during which free radicals are produced that attack the substrate molecule. It has three stages: **initiation** (the initial free radicals are produced), **propagation** (these radicals generate more radicals) and **termination** (free radicals of the same or different types combine).

$$CH_4 + 4Cl_2 \rightarrow CCl_4 + 4HCl \text{ (diffused sunlight)}$$

Initiation:

$$Cl\text{–}Cl \rightarrow Cl\bullet + Cl\bullet \text{ (sunlight)}$$

Propagation:

$$CH_4 + Cl\bullet \rightarrow \bullet CH_3 + HCl$$
$$\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

These steps are repeated to give CH_2Cl_2, $CHCl_3$ and finally CCl_4.

Termination:

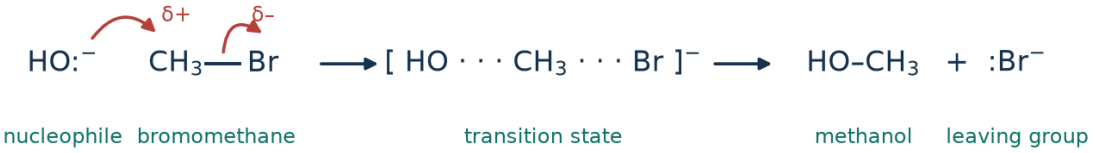
$$Cl\bullet + Cl\bullet \rightarrow Cl_2$$
$$\bullet CH_3 + Cl\bullet \rightarrow CH_3Cl$$
$$\bullet CH_3 + \bullet CH_3 \rightarrow CH_3\text{–}CH_3$$

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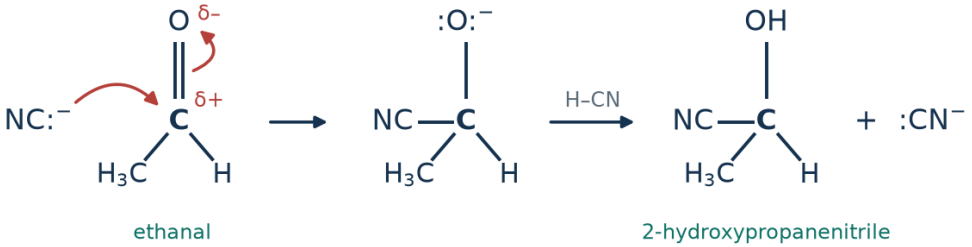
Q.41 [Long] Describe the ionic (polar) mechanisms with curly-arrow mechanisms.

The **ionic or polar mechanism** involves heterolysis, during which electrophiles and nucleophiles are produced, which may attack polar substrate molecules. It is divided into the following types.

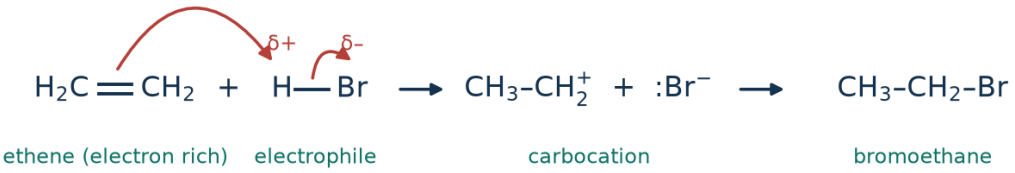
i) Nucleophilic substitution: A nucleophile attacks a saturated electrophilic carbon and substitutes another nucleophile, called the leaving group.



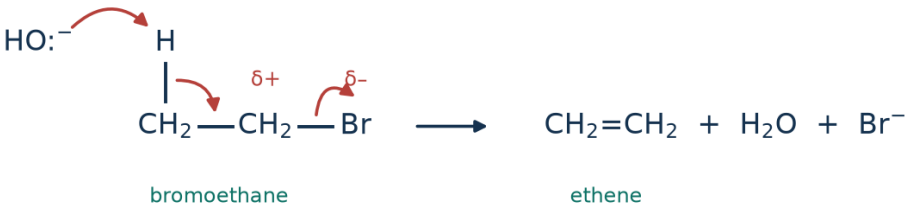
ii) Nucleophilic addition: A nucleophile attacks the electrophilic carbon of a carbonyl compound to give an addition product.



iii) Electrophilic addition: An electrophile attacks an electron-rich (nucleophilic) position such as a carbon–carbon double or triple bond.



iv) Elimination: Two atoms or groups are removed from two adjacent carbon atoms in a molecule, forming a double or triple bond.



Quick Check 19.6

Q.42 [Short] **a) Describe each of the following as electrophile, nucleophile or free radical.**

No.	Species	Type	Reason
i)	NH_3	Nucleophile	Lone pair on nitrogen can be donated
ii)	Br	Free radical	A bromine atom has an unpaired electron
iii)	H_2O	Nucleophile	Lone pairs on oxygen
iv)	NO_2^+	Electrophile	Positive ion; accepts an electron pair
v)	$Cl^{\delta+}$	Electrophile	Electron-deficient (partially positive) chlorine

Q.43 [Short] **b) Classify each of these reactions as addition, substitution, elimination or rearrangement.**

No.	Reaction	Type
(I)	$CH_3CH{=}CH_2 + HBr \rightarrow CH_3CHBrCH_3$	Addition
(II)	$CH_4 + Br_2 \rightarrow CH_3Br + HBr$	Substitution
(III)	$CH_3CH_2CHClCH_3 \rightarrow CH_3CH_2CH{=}CH_2 + HCl$	Elimination

Q.44 [Short] **c) Classify the following reactions into one (or more) of the following six categories: Substitution, addition, hydrolysis, reduction, condensation, oxidation.**

No.	Reaction	Category
(i)	$(CH_3)_2C{=}O + H_2N{-}NH_2 \rightarrow (CH_3)_2C{=}N{-}NH_2 + H_2O$	Condensation (two molecules join with loss of water)
(ii)	$CH_3CHO + 2[H] \rightarrow CH_3CH_2OH$	Reduction (gain of hydrogen); it is also an addition

19.9 ISOMERISM

Q.45 [Short] **Define isomers and isomerism.**

Two or more compounds which have the same molecular formula but different arrangements of atoms and different properties are called **isomers**, and the phenomenon is called **isomerism**.

Q.46 [Long] **Describe structural isomerism and its division into chain, positional, functional group isomerism, metamerism and tautomerism.**

Structural (constitutional) isomerism: the same groups or atoms are connected at different places in two or more molecules (different bond connectivity). Its types are:

Type	Difference between the isomers	Example
Chain isomers	Different carbon chain (straight or branched).	butane and 2-methylpropane (C_4H_{10})
Position isomers	Different position of the same functional group or substituent.	propan-1-ol and propan-2-ol (C_3H_8O)
Functional group isomers	Different functional groups.	ethanol and methoxymethane (C_2H_6O)
Metamers	Different alkyl groups on either side of the same functional group.	ethoxyethane and 1-methoxypropane ($C_4H_{10}O$)
Tautomers	Isomers in rapid equilibrium that differ in the position of a hydrogen atom and a double bond.	keto and enol forms of propanone

The textbook names these five types in a chart (structural isomerism was discussed in Chemistry 11); the one-line descriptions and examples are standard.

Q.47 [Short] What is stereoisomerism? How is it different from constitutional isomerism?

Stereoisomerism: the isomers have the same bond connectivity, but their substituents or groups have a different spatial arrangement (different configuration). Stereoisomers are divided into (a) **optical isomers** and (b) **cis-trans isomers**.

Q.48 [Short] Give the classification of isomers shown in the textbook chart.

- Isomers (same molecular formula) → **constitutional isomers** (different bond connectivity: chain, position, functional, metamers, tautomers) or **stereoisomers** (same connectivity, different spatial orientation).
- Stereoisomers interconvertible just by rotation without breaking bonds → **conformational isomers**; otherwise → **configurational isomers**.
- Configurational isomers due to restricted rotation → **geometrical isomers**; otherwise → **optical isomers**.
- Optical isomers that are non-superimposable mirror images → **enantiomers**; otherwise → **diastereomers**.

19.9.1 Stereoisomerism

Q.49 [Long] What are optical isomers? Explain optical activity.

Optical isomers are stereoisomers which have identical physical and chemical properties except their ability to rotate plane polarised light. Such isomers are said to be optically active, and this property of rotating the plane of polarisation is called **optical activity**. It is identified with plane polarised light (PPL). If the compound rotates the plane of polarised light to the right (clockwise) it is **dextrorotatory (+)**; if to the left (anticlockwise) it is **levorotatory (–)**.

A **polarimeter** is used to identify and measure quantitatively the optical activity of a compound using PPL. PPL is monochromatic light of a single wavelength which moves in

only one plane. A solution of the compound is placed before the PPL: if it rotates the PPL it is optically active, and if not, optically inactive.

Unpolarised light → polarizer → PPL → sample → plane rotated clockwise (dextrorotatory) or anticlockwise (levorotatory).

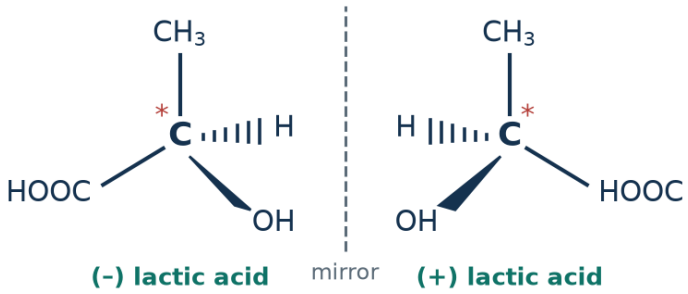
Q.50 [Long] What is the relation between optical activity and the structure of a compound? Explain chirality and the chiral carbon.

The basic requirement for a compound to be optically active is that it is **non-superimposable on its mirror image**. This condition is fulfilled by asymmetric (chiral) molecules, just as our right and left hands are non-superimposable. The word "chiral" is from the Greek *cheir*, hand; chirality is also called handedness. Molecules that are superimposable on their mirror images are **achiral** and optically inactive.

A carbon atom bearing four different atoms or groups of atoms is called a **chiral carbon** (chiral centre). A molecule with only one chiral carbon is always asymmetric and hence optically active. If there is more than one chiral carbon, the molecule may or may not be asymmetric and optically active.

Q.51 [Long] What are enantiomers? Explain with the example of lactic acid.

Lactic acid, CH_3–CH(OH)–COOH, has one chiral carbon bonded to four different groups: CH_3, H, OH and COOH. Its three-dimensional structure has two stereoisomers which are non-superimposable mirror images of each other, called **enantiomers** ("enantio" means opposite). Enantiomers have identical physical and chemical properties, but they rotate the plane of polarised light in opposite directions to the same extent: one is dextrorotatory (+), the other levorotatory (–).



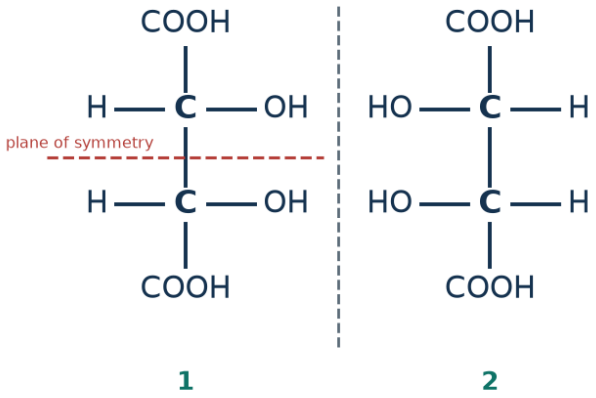
Enantiomers of lactic acid (= chiral carbon)*

Q.52 [Short] What is a racemic mixture?

When 50% dextrorotatory and 50% levorotatory enantiomers are present, they form an optically inactive mixture called a **racemic mixture**. It is symbolised by (±) in front of the name. Thus lactic acid exists in two optically active forms, (+) lactic acid and (–) lactic acid, and an inactive racemic mixture (±).

Q.53 [Long] What are meso isomers? Why are they optically inactive?

Stereoisomers which contain at least two similar chiral carbon atoms and also possess a **plane of symmetry** are optically inactive. The plane of symmetry divides the molecule into two equal halves, one half being the mirror image of the other. The two halves rotate the plane of polarised light in opposite directions, so the rotation of one half is internally compensated (cancelled) by the other half. Such stereoisomers are called **meso forms**. Tartaric acid is the classic example.



Structures 1 and 2 are identical and represent the meso form of tartaric acid

Q.54 [Short] How is the maximum number of optical isomers of a compound calculated?

The maximum number of optical isomers of an optically active compound is given by $\mathbf{2^n}$, where n is the number of chiral carbon atoms. This number is reduced if there are two or more similar chiral carbon atoms.

Q.55 [Short] Differentiate between enantiomers and diastereomers.

Enantiomers are optical isomers that are non-superimposable mirror images of each other. Diastereomers are configurational stereoisomers that are not mirror images of each other.

Quick Check 19.7

Q.56 [Short] Q1: Classify the following objects as chiral or achiral. (a) Pencil (b) Book (c) Shoe (d) Cup (e) Ear (f) Foot (g) Egg (h) Spoon

Chiral (not superimposable on the mirror image): shoe, ear, foot. **Achiral** (have a plane of symmetry): pencil, cup, egg, spoon and a plain book.

A book is achiral only as a plain object; if its printing and the side on which it opens are considered, it is chiral.

Q.57 [Short] **Q2: Glucose and fructose are two monosaccharides. i. How many chiral carbon atoms are there in glucose and fructose? ii. Calculate the possible number of optical isomers of glucose and fructose. iii. Is there any meso form in glucose or fructose? Give reason.**

i. In the open chain forms, glucose, $CH_2OH–(CHOH)_4–CHO$, has **4** chiral carbon atoms (C-2 to C-5); fructose, $CH_2OH–CO–(CHOH)_3–CH_2OH$, has **3** (C-3 to C-5).

ii. Number of optical isomers = 2^n. Glucose: 2^4 = **16**. Fructose: 2^3 = **8**.

iii. No. A meso form needs similar chiral carbons and a plane of symmetry. In glucose the two ends are different (–CHO and $–CH_2OH$), and in fructose the C=O group is at C-2, so neither molecule can be divided into two mirror-image halves.

The open chain structures of glucose and fructose are not given in this chapter; they are used here from general knowledge.

19.9.1 Cis-trans Isomerism

The textbook numbers both the Stereoisomerism and the Cis-trans Isomerism headings as 19.9.1.

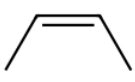
Q.58 [Long] **What is cis-trans isomerism? Explain its origin in terms of restricted rotation due to the π bond.**

Stereoisomers having a different spatial arrangement of atoms due to **restricted rotation about a double bond** in open chain compounds are called **cis-trans isomers**.

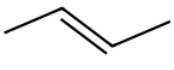
Free rotation about a single bond: in butane the C–C single bond between C-2 and C-3 rotates freely, so the methyl groups also rotate and form I changes into form II and back. Forms I and II are **conformational isomers**: they are interconvertible by free rotation and cannot be separated at room temperature.

Restricted rotation about a double bond: in but-2-ene, $CH_3–CH=CH–CH_3$, carbon atoms 2 and 3 are sp^2 hybridized. The C=C double bond consists of a σ bond (linear overlap of sp^2 hybrid orbitals) and a π bond (parallel overlap of p orbitals). The π bond locks the molecule in one position: C-2, C-3 and all the groups attached to them lie in the same plane and their positions in space are fixed. For free rotation the π bond must first be broken, which requires 263 kJ/mol; this is not possible at room temperature. This restriction of rotation is responsible for cis-trans isomerism.

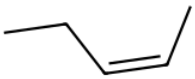
The **cis** isomer has the two similar groups on the same side of the double bond; the **trans** isomer has them on opposite sides.



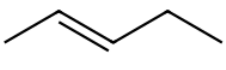
cis-but-2-ene



trans-but-2-ene



cis-pent-2-ene



trans-pent-2-ene

Q.59 [Short] What is the necessary condition for cis-trans isomerism? Why do but-1-ene and 2,3-dimethylbut-2-ene not show it?

Condition: for an open chain compound the necessary and sufficient condition is that it has a double bond and that two different groups are attached to each of the doubly bonded carbon atoms. Cis-trans isomerism is not possible if one of the unsaturated carbon atoms is bonded to two identical groups, e.g., but-1-ene (CH_2=CH–CH_2–CH_3) and 2,3-dimethylbut-2-ene, $(CH_3)_2C=C(CH_3)_2$.

Q.60 [Short] How do cis and trans isomers differ in their properties?

Cis-trans isomers have different physical and chemical properties. Trans isomers are more stable than the corresponding cis isomers and hence differ in chemical reactivity. They also differ in physical properties such as melting point, boiling point, dipole moment, density and heat of vapourisation.

Q.61 [Short] Differentiate between conformational isomers and cis-trans isomers.

Conformational isomers	Cis-trans isomers
Arise from free rotation about a C–C single bond.	Arise from restricted rotation about a C=C double bond.
Interconvertible; cannot be separated at room temperature.	Not interconvertible without breaking the π bond (263 kJ/mol); can be separated.
Forms I and II of butane.	cis- and trans-but-2-ene.

19.10 STEREOCHEMISTRY IN DRUG DEVELOPMENT

Q.62 [Long] Explain the significance of chirality in drug development, using thalidomide as an example.

In the synthesis of pharmaceutical products chirality is a critical feature for safety and efficacy. Biological systems are made of chiral building blocks, and drug receptors act as highly specific “chiral locks”. So the two enantiomers of a chiral drug often have vastly different potentials and profiles.

Thalidomide, a tranquilliser, was originally distributed as a racemic mixture. It seemed a very successful, non-toxic sedative with no known side-effects. In the early 1960s it was discovered that women who had been prescribed it to treat nausea (“morning sickness”) in early pregnancy had produced badly deformed babies, and the drug was immediately withdrawn. Research showed that the (+) isomer was an effective and safe tranquilliser, while the (–) isomer damaged the fetus. The problem cannot be solved by giving the pure (+) isomer, because it racemises rapidly as soon as it enters the blood stream: within 10 minutes racemisation is over 50% complete. Thalidomide will never be marketed again, although chirality-stable analogues are being investigated as possible anti-inflammatory drugs.

Q.63 [Short] Why must racemic mixtures of drugs be separated, and how can a single optical isomer be produced?

The two enantiomers of a chiral drug can have very different biological activity: one may be the useful medicine while the other is inactive or harmful (as with thalidomide). So the racemic mixture has to be separated to obtain the useful isomer, which is difficult and costly. A better way is to use a **chiral catalyst** in the synthesis, which produces only the single pure optical isomer that is required.

The use of chiral catalysts is listed in the learning outcomes of the chapter but is not explained in its text.

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